警告Warning

加热器在通电以后，请不要接触，以防触电。
After the heater in electricity, please don't touch to prevent electric shock.

在检查、进行设备维护的时候，请切断电源。
For maintenance or check, please cut off the power supply.

注意Note

对于该设备，请使用指定的电压。
For the device, please use the specified voltage.

设备在运行时请勿接触或靠近可燃性物体，避免引起火灾。
The device in runtime does not touch or near flammable objects, avoid causing a fire.

设备在运转过程中如发生声音异常、过热时请立即停止该设备。
Equipment in operation process， such as the occurrence of abnormal sound、overheating and so on, please immediately stop the equipment.

设备运行时泵体内绝对不能暴露大气。
The pump inside is not exposed to air during device operation.

必须对该设备进行定期检查。
Must be regularly checking the equipment.

请在有使用经验人员的指导下操作该设备。
Please operate the equipment have had experience in the use of personnel under the guidance.

承诺Commitment

我公司的产品在出厂前已经过严密测试。如果在使用中出现故障、或者在运输过程中设备损伤，请和我公司联系。
Our company's products have been strictly tested before delivery. If a failure occurs in use or in the transportation process or equipment damage, please contact us.

自您购买产品之日起，我公司将为您所购设备进行一年期的“三包”服务。
Since the date of the purchasing, our company will provide you the purchase equipment for one year the "Three Guarantees" service.

凡属下列情况之一，将不在“三包”范围以内：
Any of the following circumstances, will not be within "Three Guarantees"

- 未经授权对设备进行改造
  Unauthorized equipment renovation

- 用户保管或使用不当
  Improper use or storage

- 属于用户其它原因造成的损坏
  Damaged · belonging to the user other causes

- 因不可抗力因素造成的损坏
  For damage caused by irresistible factors

免责声明Disclaimer

当用户严格遵照本说明书中的规定安装和使用时，我公司的设备是安全、方便和有效的。When the user is strictly in accordance with the provisions of the installation and use, our company's equipment is safe, convenient and effective.

由于用户没有仔细阅读说明书、或未按照使用说明书的要求操作而造成的任何伤害和损失，我公司将不承担任何责任。
Because the user has not read the instructions carefully, or not in accordance with the requirements of the specification and cause any harm or loss, our company will not bear any responsibility.

本说明书仅作为信息使用，如遇改版，恕不另行通知。若本说明书与实际产品有所出入，本公司拥有最终解释权。
The specification is just for referring only. If in case of revision we will no notice in prefer. If there is difference between the product and the instructions, we have the right of final interpretation.

概述Summary

主要用途及适用范围
The main purpose and application

高真空金属油扩散泵（以下简称扩散泵）适用于电子工业、化学工业、真空冶炼、原子能工业等领域，对被抽容器进行抽气，使之达到高真空（可达10-5Pa）目的。
High vacuum metal oil diffusion pump (hereinafter referred to as the diffusion pump) is used in electronic industry, chemical industry, vacuum metallurgy, atomic energy industry and other fields, in order to take out the air by pumping the container, so as to achieve the purpose of high vacuum (up to 10-5Pa).

扩散泵适用于抽除高真空系统中对泵油无严重危害的气体。
Diffusion pumps for pumping gas without serious damage to pump oil in high vacuum system.

型号组成及其代表意义
Composition and represents the meaning of type

扩散泵由K型和 KT型组成
Diffusion pump is composed of K type and KT type

K型扩散泵的泵壳为直腔，KT型扩散泵的泵壳为凸腔
The pump casing of K diffusion pump is straight cavity while the pump casing of KT diffusion pump is convex cavity

K(油扩散泵diffusion pump)**—100(**进气口直径inlet diameter)

K(油扩散泵diffusion pump) **T(**泵壳凸腔形convex cavity)**—200(**进气口直径inlet diameter)

推荐使用环境 recommended the use of the environment

环境温度environment: temperature	10～35℃
湿度moisture	≤85％
电源supply voltage	380v ±5％
冷却水水压cooling water pressure water pressure	0.2～0.4Mpa；
冷却水水质cooling water quality	工业自来水industrial water
进水水温要求inlet water temperature	4 ～20℃

工作原理及结构特征the work principle and structure feature

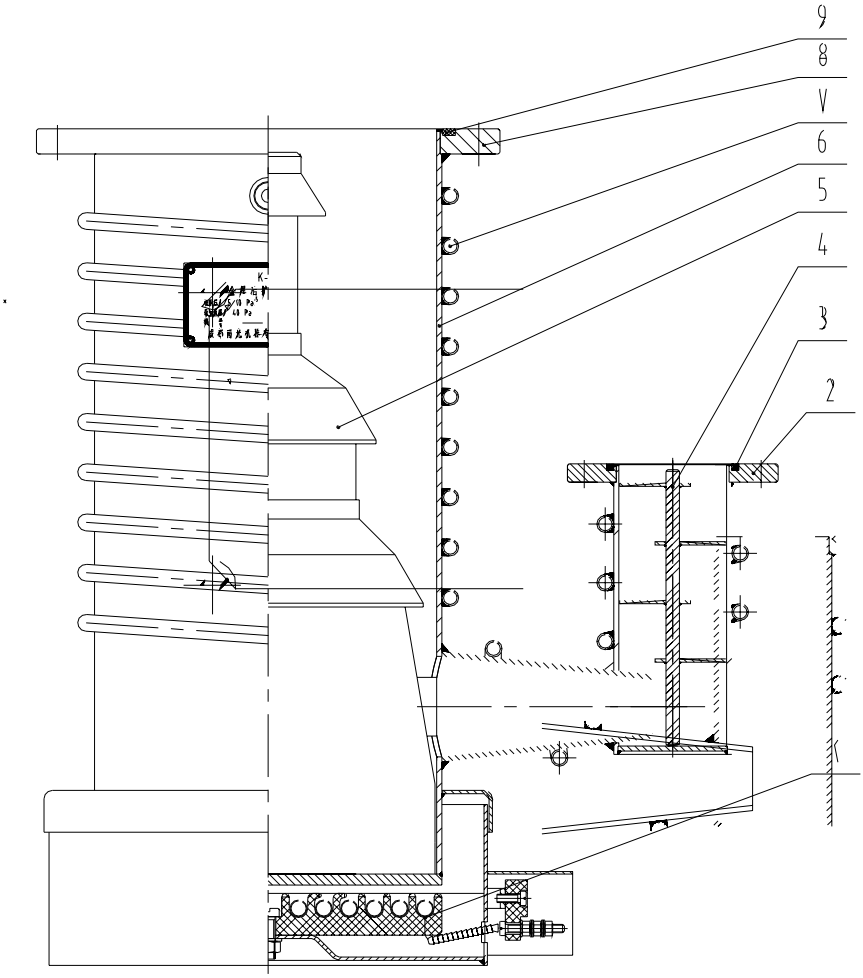
工作原理the work principle

扩散泵油锅中的泵油在真空中（＜10Pa）加热到沸腾温度，产生大量的油蒸汽，油蒸汽经导流管由各级喷嘴定向高速喷出。
Diffusion pump oil in vacuum (< 10Pa) is heated to boiling temperature, produce large amounts of oil vapor, the vapor passes through the guide tube directed by various high speed from the nozzle.

由于扩散泵进气口附近被抽气体的分压力高于蒸汽流中该气体的分压力，所以被抽气体分子就不断地扩散到蒸汽流中。油蒸汽撞击被抽气体分子，使被抽气体分子沿蒸汽流方向高速运动。气体分子碰到泵壁又反射回来，再受到蒸汽流的碰撞而重新沿蒸汽流方向流向泵壁。
Because the pressure of diffusion pump nearby inlet is higher than that of the vapor stream, so the pumped gas molecules will continue to spread to the steam flow. Oil vapor impact pumped gas molecules, the pumped gas molecules along the vapor flow direction high speed movement. The molecules of a gas meet wall of the pump and back again, by the collision of steam flow and steam flow direction flow along the wall of the pump.

经过几次碰撞后，气体分子被压缩到低真空端，再由下几级喷嘴喷出的蒸汽流进行压缩，最后由前级真空泵抽走。而油蒸汽在冷却的泵壁上被冷凝后又返回到油锅中重新被加热，如此循环工作。
After several times after the collision, the molecules of a gas is compressed to a low vacuum side, again by a few class of steam spray compression, and finally by the backing pump away. While the oil vapor in the wall of the pump cooling is condensed and returned to the pan to be heated, then the cycle to work.

结构特征structural features



1. 加热器heater
2. 排气口法兰outlet flange
3. 排气口密封圈outlet sealing ring
4. 油挡oil shelf
5. 泵芯pump core
6. 泵壳pump shell
7. 冷却水管cooling water pipe
8. 进气口法兰 inlet flange
9. 进气口密封圈 inlet sealing ring

扩散泵主要由泵壳、泵芯、蒸发锅，加热器、冷却水管等组成。KT型泵是将泵壳在一级喷嘴与二级喷嘴之间做成凸腔形，以增加抽气速率。
Diffusion pump is mainly made by the pump shell; pump core, evaporation boiler heater and cooling water pipe. KT pump is a type of oil pump which the convex cavity made between between the first nozzle and the second nozzle, in order to increase the pumping rate.

泵壳主要供装置泵芯，隔绝大气，冷凝油蒸汽用。内表面经抛光处理。泵芯由多级喷嘴及蒸汽导管组成，并将其固定在泵壳内。
Pump shell is mainly used for put pump core, isolating atmosphere and condensing

oil vapor. The inner surface after polishing. Pump core is composed of multi nozzle and a steam conduit, and is fixed in the pump shell.

技术参数及外形连接尺寸Technical parameters and connecting flange size

主要参数main parameters

型号 Types	入口直径entrance diameter (mm)	抽速pumping speed (L×S-1)	极限压力extreme pressure (Pa)	最大排气压力max. Exh. Pre. (Pa)	注入油量injection oil(L)	加热功率heating power (KW)	冷却水流量cooling water flow (L×h-1)	推荐前级抽速recommended pre pumping speed(L×S-1)	输入电压input v. (V)
K-100	100	300	5×10-5	40	0.11	0.65	100	4	220
K-100B	100	300	5×10-5	40	0.11	0.65	100	4	220
K-150	150	800	5×10-5	40	0.3	1	150	8	220
K-200	200	1500	5×10-5	40	0.5	1.5	250	15	220
K-300	300	3300	5×10-5	40	1.3	2.1	300	30	220
K-400	400	5300	5×10-5	40	2.5	4.8	600	70	380
K-630	630	13000	5×10-5	40	6	8.4	850	150	380
K-630A	630	14000	5×10-5	40	8	14.4	850	150	380
K-800	800	25000	5×10-5	40	15	19.8	1200	240	380
K-1000	1000	40000	5×10-5	40	22.5	26.4	1600	300	380
TK-150	150	1000	5×10-5	40	0.3	1	150	8	220
TK-300	300	4000	5×10-5	40	1.3	2.4	300	30	220
KT-200	200	2000	5×10-5	40	0.55	1.8	250	15	220
KT-400	400	7000	5×10-5	40	3.5	4.8	600	70	380
KT-500C	500	11000	5×10-5	40	5	6	700	70	380

备注Remarks

K-630A、K-800、K-1000采用加热棒加热，每根加热棒的工作电压为220 V
K-630A, K-800 and K-1000 by heating rod, the working voltage of each heating rod is 220 V.

K-100和K-100B采用加热盘加热
K-100 and K-100B by heating plate heating,

其余均采用电炉加热
The rest types using electric heating;

外形及连接尺寸external and connecting dimensions

外形及连接尺寸由图2及下表给出
External and connecting dimensions from Fig. 2 and given in the table below

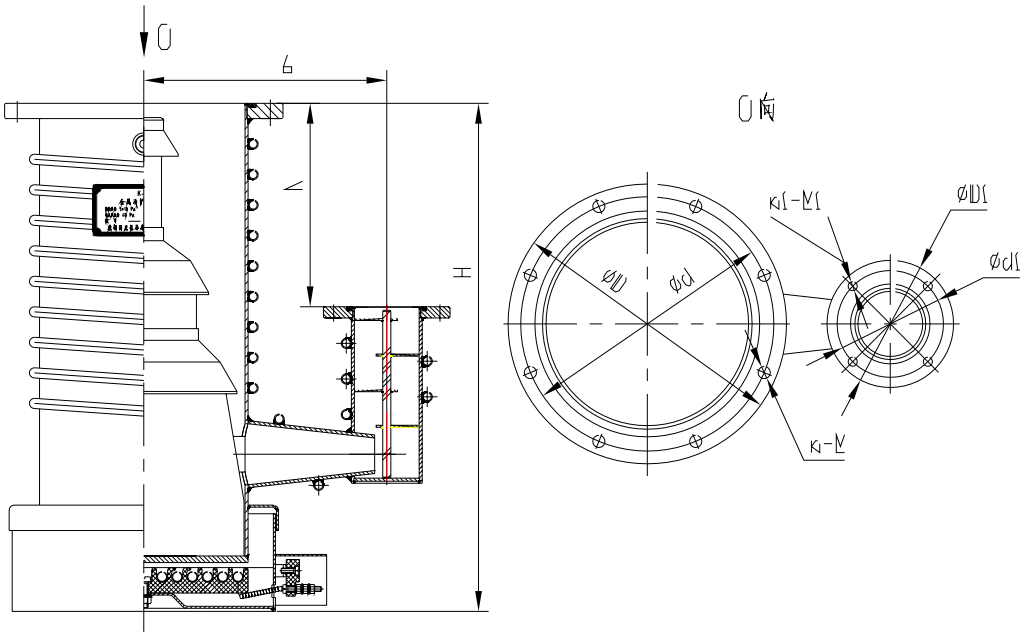


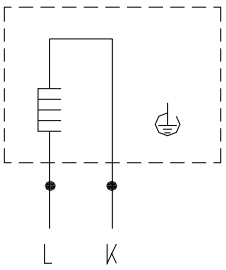
图2

外形连接尺寸connecting flange size

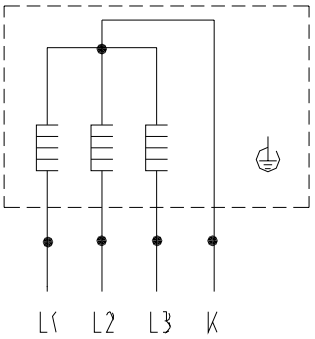
型号 type	D	d	D1	d1	H	A	L	n-M	n1-M1
K-100	170	145	78	64	315	130	125	4-M10	4-M6
K-100B	170	145	78	64	315	130	125	4-M10	4-M6
K-150	220	195	110	90	470	180	165	8-M10	4-M8
K-200	275	250	125	105	515	240	200	8-M10	4-M8
K-300	380	350	145	125	690	330	280	8-M12	4-M8
K-400	500	465	170	145	840	440	360	8-M16	4-M10
K-630	750	720	225	200	1110	520	500	20-M12	8-M10
K-630A	750	720	165	145	1260	510	500	20-M12	8-M8
K-800	920	890	225	200	1560	600	965	24-M12	8-M10
K-1000	1120	1090	225	200	2000	815	1275	32-M12	8-M10
TK-150	220	195	110	90	470	180	165	8-M10	4-M8
TK-300	380	350	145	125	720	330	280	8-M12	4-M8
KT-200	275	250	125	105	550	240	200	8-M10	4-M8

KT-400	510	480	165	145	900	440	360	16-M12	8-M8
KT-500C	610	580	200	175	1110	500	500	16-M12	8-M10

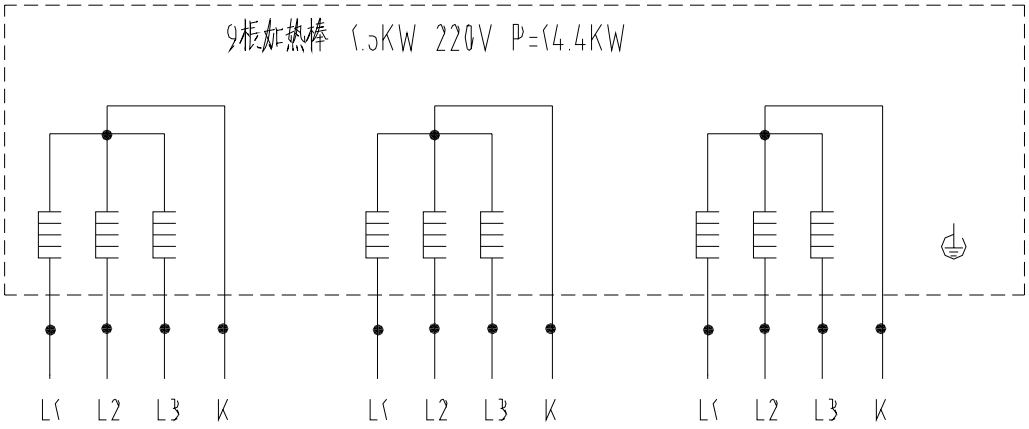
加热器接线原理图the heater wiring diagram



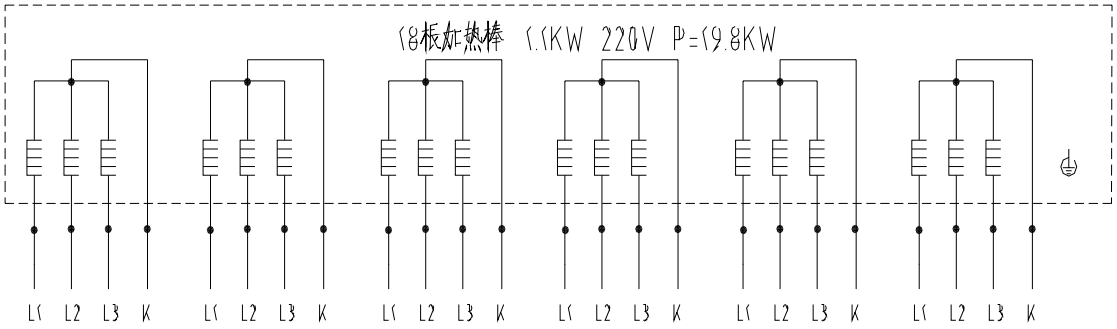
适用于K-100、K-100B、K-150
K-200、K-300、TK-150、
KT-200、TK-300扩散泵



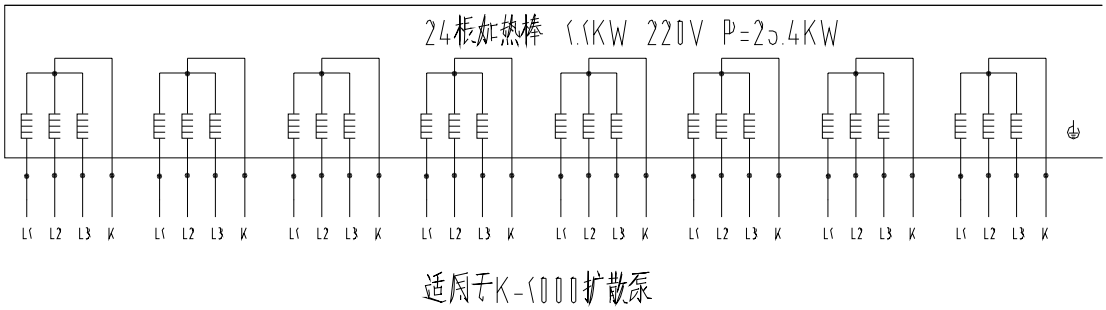
适用于K-400、KT-400
KT-500C、K-530扩散泵



适用于K-530L扩散泵



适用于K-800扩散泵



开箱检查check

打开包装箱先确认以下内容
Open the packing box to confirm the following contents

是否是定货所规定的型号
Is it the prescribed order model?

运输过程中是否有损坏等情况
Whether there is damage during transport process

按装箱单检查合格证、密封圈等附件是否附带
According to the packing list and inspection certificate, seals and other accessories are included

如果有产品外观破损、零件缺少等情况请立即和我公司联系

If there is the appearance of the product damage or parts missing, please contact us

安装准备installation preparation

先将放在进气口、排气口等部位的保护盖板取下
First of all, remove the cover on the part of air inlet and outlet

确认泵芯、冷帽、泵体是否有损坏、外观是否正常
Make sure pump core, cold cap, pump body is damaged or not

我公司的扩散泵出厂前均已进行了性能测试，并装有充足的泵油。如果发现扩散泵因其它原因造成油量不足或无油时，请按该设备的要求加入扩散泵油。

We have tested the diffusion pump after delivery. If it is found that the diffusion pump due to insufficient oil or no oil, please join in the equipment of oil as requirement

提示：扩散泵存放时间超过12个月时，应检查密封圈是否变坏，泵内零部件和泵油是否变坏。建议更换密封圈和泵油。
Warm reminder: If the store time of diffusion pump is more than 12 month, please check the sealing ring, pump parts and pump oil. Recommend replacing sealing ring and pump oil.

⚠ **注意**在没有加入扩散泵油的情况下加热，可能会造成加热器损毁、泵芯变形甚至损坏。

⚠ **Note:** Heating the pump in the absence of pump oil may cause damage to the pump itself.

请检仔细查加热器。加热器的引线、电极等不能与机壳接触以免短路或漏电。
Please check the heater carefully. The heater wire, electrode can not contact with the shell so as to avoid short circuit or leakage.

接好电路（参照加热器接线图）并进行通电试验。
Connect the circuit (reference to heater wiring diagram) and electric test.

⚠ 警告：加热器请接地保护
⚠ Warning: Heating after grounding protection

在法兰、密封圈等部位用酒精等挥发性溶剂等进行清洁，并进行表面干燥处理
Please use alcohol and other volatile solvent to clean the flange and sealing ring and other parts, then dry surface treatment.

用适量的真空脂均匀涂敷于密封圈表面。
Use uniform amount of vacuum grease coating on the sealing ring surface.

安装 installation

管道连接
pipe connection

在安装法兰的时候，注意保证两个法兰平行，确认密封圈没有脱糟的现象。
When installing flange, pay attention to ensure that the two flanges in parallel, that the sealing ring is not off.

冷却水连接
Cooling water connection

扩散泵必须通冷却水冷却。冷却水进、出应保证上进下出，反接会影响泵的性能。
The diffusion pump must be cooled by cooling water-Cooling water inlet, outlet shall ensure that top in and bottom out, and reverse connection will affect the pump performance

泵壳要求分段接入冷却水时，每段应提供独立的水冷通路，串联供水会影响泵的冷却效果从而影响泵的性能。
Pump shell segment access cooling water, water cooling path of each section should provide independent. Tandem water will affect the pump cooling effect which affects the pump performance

水冷挡帽请提供独立的水冷通路，冷却不足会造成泵返油率增加。
Please provide independent water channel to water retaining cap, insufficient cooling water will cause increased rates of return oil.

电气配线
Electrical wiring

请按照规定的电压接入电源，并在正式通电前再次确认电压是否为我们规定的电压值。Please follow the prescribed voltage access to power, and in the formal electrifying before confirming whether the voltage is the voltage we specified value again.

检漏
Leak detection

扩散泵安装完成以后，在泵加热器正式加热前，请对整个系统进行真空检漏。在具备条件的情况下，请使氦质谱检漏仪对整个系统的真空状况进行氦质谱检漏。
After the installation is complete, before the pump heating, please do vacuum leak

detection to the whole system. In proper conditions, please use helium mass spectrometer leak detector to the whole system.

运转
Operation

启动
Boot

接通冷却水，确认冷却水的流量是否达到规定的要求。
Connected to the cooling water, to confirm whether the flow of cooling water requirements.

启动前级泵，当扩散泵泵口压力≤10Pa后，才能启动扩散泵的加热电源。
Before start the pump, when the diffusion pump outlet pressure below 10Pa, can start the heating power of the diffusion pump.

泵加热40～80分钟后，就进入正常工作状态。
After pump heating 40 ~ 80 minutes, can enter the normal working state.

停止
Stop

先关闭高真空阀门，然后切断扩散泵加热电源，前级泵继续工作。
Turn off the high vacuum valve first, and then cut off the power supply of the diffusion pump, the backing pump continue to work.

当泵体内部温度降到≤60℃的时候（用手可以接触的状态），可以停止前级真空泵。wh en the temperature inside the pump body to ≤ 60 ℃(may touch by hand), can stop the backing vacuum pump.

关闭冷却水。
Close cooling water.

注意事项
The matters needing attention

向泵体内放气时，必须在泵油温度≤60℃时进行。油温度过高，会造成泵油氧化，导致泵性能下降。
To vent the pump, must be in the pump oil temperature ≤ 60 ℃. The oil temperature is too high will cause the pump oil oxidation, resulting in pump performance drop.

放气时请从高真空端放气，避免损坏泵芯。如果不能从高真空端放气时，请保证小流量缓慢放气。
Please release gas from the high vacuum side, to avoid damage to the pump core. If not from high vacuum to end deflation, please ensure that the small flow slowly deflated.

扩散泵在正常运行中遇意外停电、停水时，请按以下步骤处理。
When the accident of power cut and stops of the water supply, please according to the following steps:

切断电源
Cut off the power supply

立即关闭高真空阀门和前级真空阀门
Immediately close the high vacuum valve and the backing vacuum pump valve

将电炉从泵底卸下，并移开。（注意安全，小心烫伤！！！）
Unloaded and move away the furnace heater (Caution! burn!!!)

向泵底部（油锅部位）淋水（应备有高位水箱）。注意：水蒸气烫人！
Pouring water to the bottom of the pump (with water tank in top). Note: hot water vapor burn!

在停止作业后，扩散泵短期内不使用时，请将泵水冷管道中的水排除管道以外，同时用压缩空气将残余水分吹干，防止水冷管道锈蚀。
When stop use the pump in short term, please drain the water away and use of compressed air to blow the residual moisture inside to prevent water pipeline corrosion.

扩散泵工作一段时间后，若发现真空度变坏，或抽速降低使系统恢复真空时间太长时，必须对泵清洗，更换泵油。
The pump work after a period of time, if it is found that the degree of vacuum pumping speed reducing, or restore the system vacuum time is too long, it need to pump cleaning and replacement of oil pump.

扩散泵应存放在温度为10℃～35℃、湿度不超过85％的通风良好的室内。
The diffusion pump should be stored at a temperature of 10 ℃ ~ 35 ℃, relative humidity of not more than 85%ventilated room.

常见故障分析与排除
Analysis and troubleshooting of common failures

扩散泵的常见故障原因及排除方法
Common failure and troubleshooting of the diffusion pump

故障 **Fault**	**原因** **Elimination**	**排除方法** **Method**
扩散泵不起作用 Diffusion pump not work	系统漏气，超过允许漏率system leakage, exceed the allowable leakage rate	进行检漏，找出漏孔并排除 leak detection and find out
	加热器不起作用 heater doesn't work	检查电源是否有电，加热器是否断路 check whether the heater is broken
	油加热不足，油温低 oil heating shortage, low oil temperature	检查电源电压是否符合规定，加热器电阻丝是否烧断，炉盘离锅底距离是否适当； check the power supply voltage, the resistance heater wire, or the distance between the bottom of the pot and wire pot
	前级压力过大 level pressure before check	检查前级是否漏气，前级泵配置是否合理 Check the leak of backing pump or the configuration of backing pump
	喷嘴松动 nozzle loose	重新对喷嘴进行紧固 Fastening
抽气速率过低 Pumping rate too low	油温过低 oil temperature too low	检查供电电压及加热器是否符合规定；加热器是否烧断，并消除或更换 checking the power supply and the heater, if the heater is burnt, please remove and replace
	泵内泵油过少 oil too little	装入新油 Replace new oil
	泵芯安装不正确 pump core is not installed correctly	检查泵芯有无松动、偏斜及喷嘴间隙是否正确，并重新调整。 Check the pump core has not loosening, or deflection, and the nozzle clearance is correct, and re-adjusts.
极限压力达不到要求 Ultimate pressure is not up to the requirements	真空系统漏气 vacuum system leakage	对系统检漏，找出漏孔，并消除 leak detection, find the leak and eliminate
	真空系统或泵内不清洁 Vacuum system or pump is not clean	保证真空卫生，拆开清洗干净 Wash and keep clean

	泵油氧化、裂化 oil oxidation or cracking	换新油 Replace new oil
	扩散泵冷却不良 diffusion pump cooling bad	检查水温及水路是否畅通，水的流量是否满足，并消除； check water temperature and the water flow
	油量过多或过少 oil quantity is too much or too little	增加或减少泵油； increase or decrease the oil

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